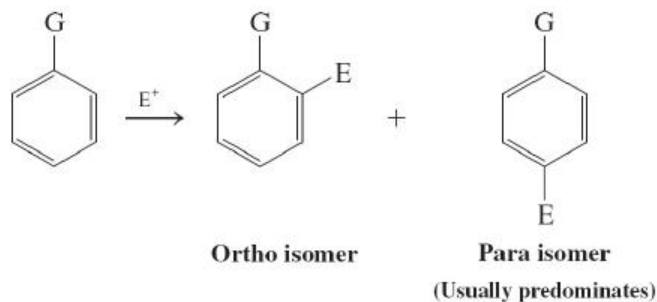


New Reactions

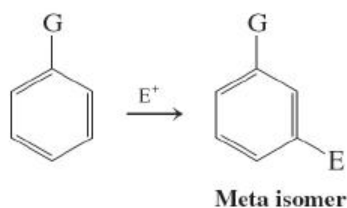
Electrophilic Substitution of Substituted Benzenes

1. Ortho- and Para-Directing Groups ([Sections 16-1](#) through [16-3](#))



G = NH_2 , OH; strongly activating
= NHCOR , OR; moderately activating
= alkyl, aryl; weakly activating
= halogen; weakly deactivating

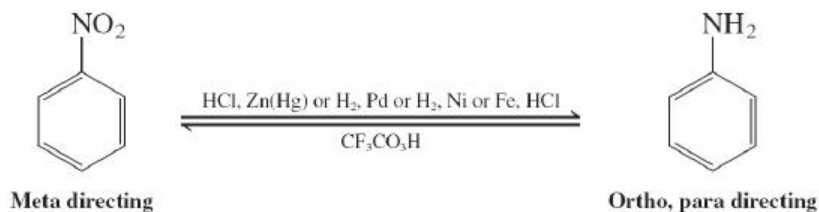
2. Meta-Directing Groups ([Sections 16-1](#) through [16-3](#))



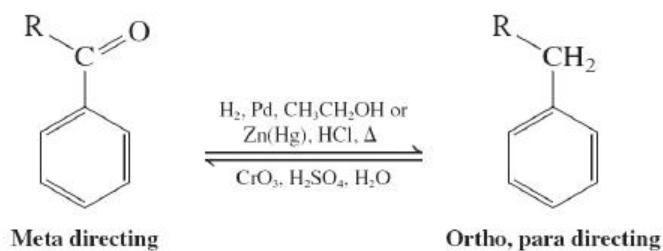
G = $\text{N}^+(\text{CH}_3)_3$, NO_2 , CF_3 , $\text{C}\equiv\text{N}$, SO_3H ; very strongly deactivating
= CHO, COR, COOH, COOR, CONH_2 ; strongly deactivating

Synthetic Planning: Switching and Blocking of Directing Power

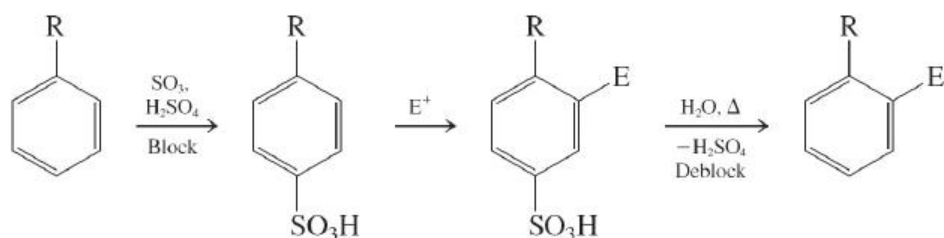
3. Interconversion of Nitro and Amino Groups ([Section 16-5](#))



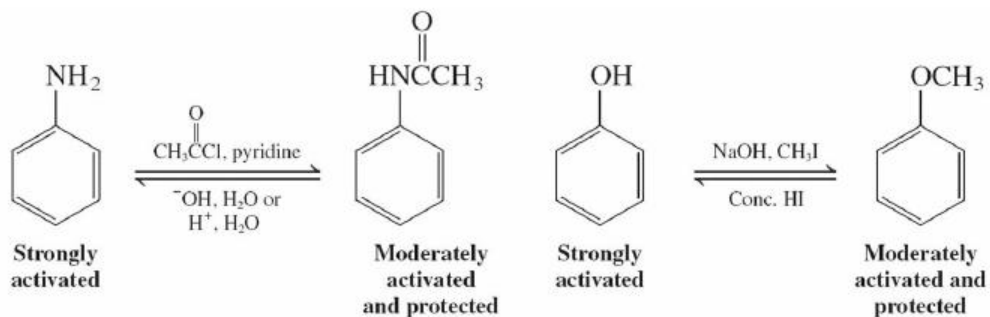
4. Interconversion of Acyl and Alkyl ([Section 16-5](#))



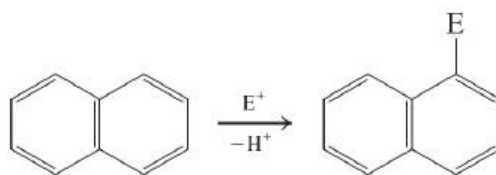
5. Blocking by Sulfonation ([Section 16-5](#))



6. Moderation of Strong Activators by Protection ([Section 16-5](#))

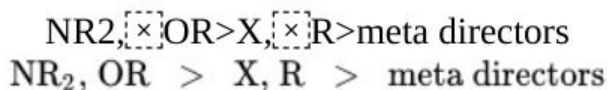


7. Electrophilic Aromatic Substitution of Naphthalene ([Section 16-6](#))



Important Concepts

1. Substituents on the benzene ring can be divided into two classes: those that **activate** the ring by **electron donation** and those that **deactivate** it by **electron withdrawal**. The mechanisms of donation and withdrawal are based on **induction** and **hyperconjugation** or **resonance**. These effects may operate simultaneously to either reinforce or oppose each other. Amino and alkoxy substituents are strongly activating, alkyl and phenyl groups weakly so; nitro, trifluoromethyl, sulfonyl, oxo, nitrile, and cationic groups are strongly deactivating, whereas halogens are weakly so.
2. **Activators** direct electrophiles **ortho** and **para**; **deactivators** direct **meta**, although at a much lower rate. The exceptions are the **halogens**, which deactivate but direct **ortho** and **para**.
3. When there are several substituents, the strongest activator (or weakest deactivator) controls the regioselectivity of attack, the extent of control decreasing in the following order:



4. Strategies for the **synthesis** of highly substituted benzenes rely on the **directing power** of the substituents, the synthetic ability to **change** the **sense of direction** of these substituents by chemical manipulation, and the use of **blocking** and **protecting** groups.
5. **Naphthalene** undergoes preferred electrophilic substitution at **C1** because of the relative stability of the intermediate carbocation.
6. Electron-donating substituents on one of the naphthalene rings direct electrophiles to the same ring, ortho and para. Electron-withdrawing substituents direct electrophiles away from that ring; substitution is mainly at **C5** and **C8**.
7. The actual carcinogen derived from benzo[a]pyrene appears to be an oxacyclopropanediol in which **C7** and **C8** bear hydroxy groups and **C9** and **C10** are bridged by oxygen. This molecule alkylates one

of the nitrogens of one of the DNA bases, thus causing mutations.